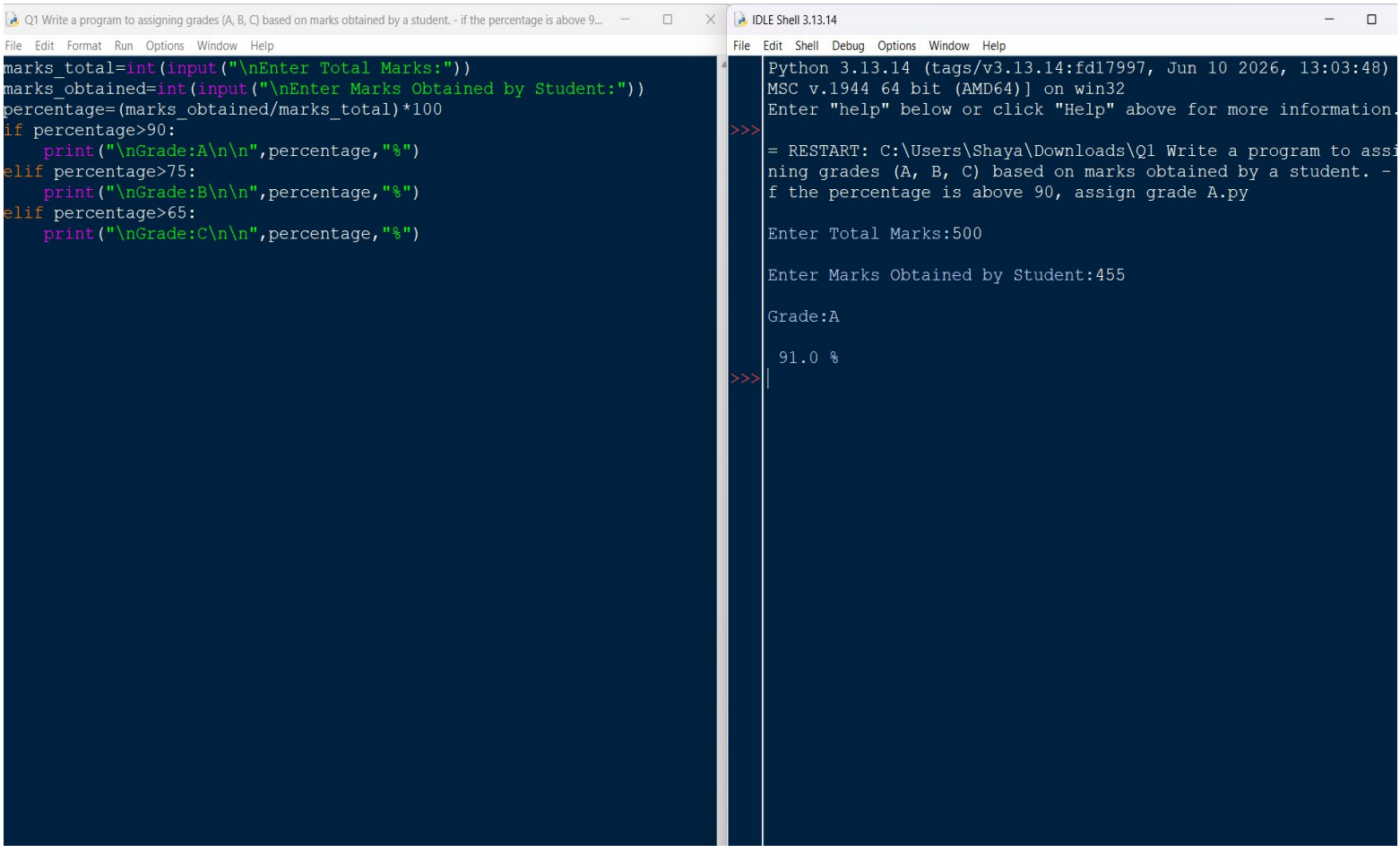
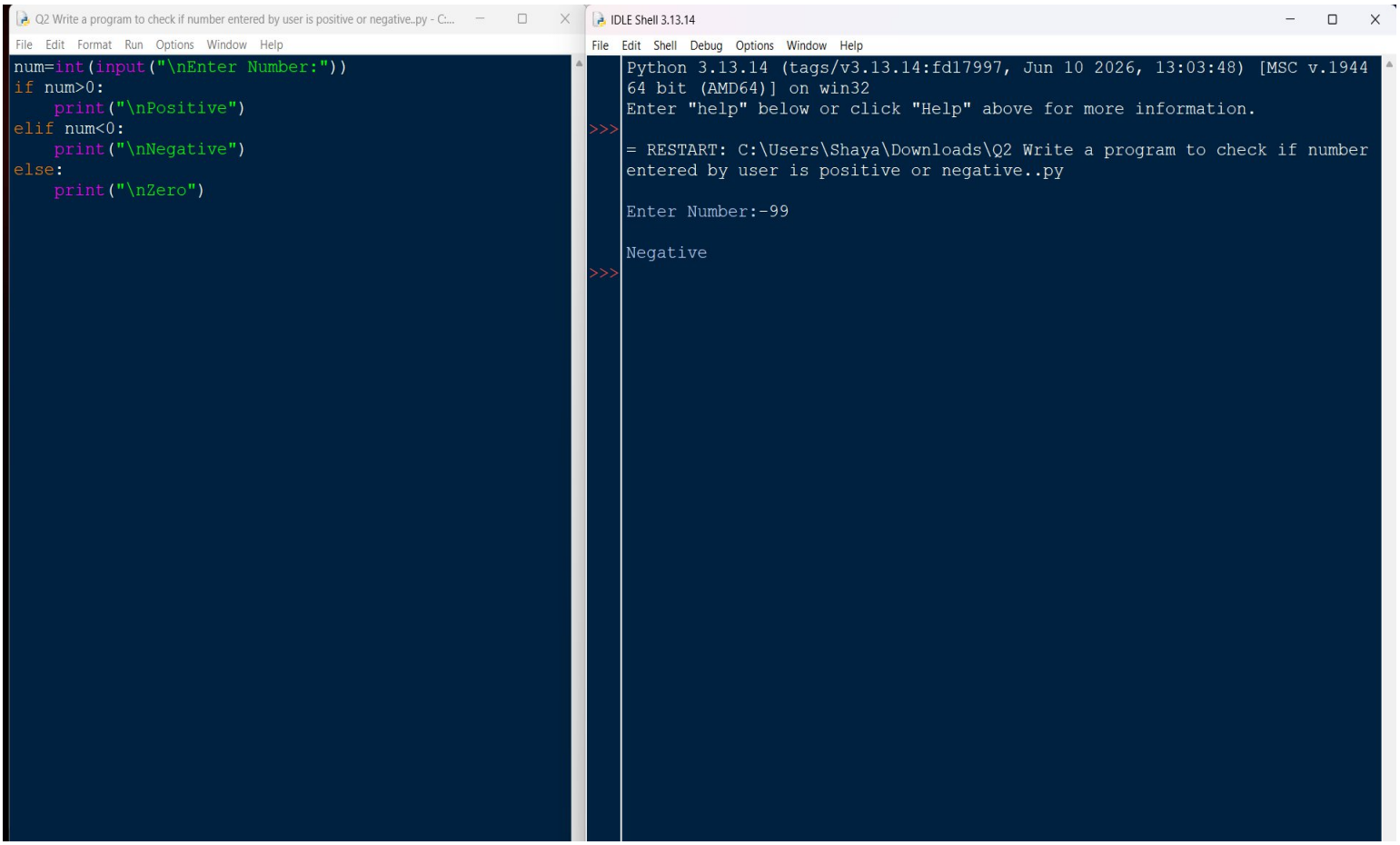
**Q1. Write a program to assigning grades (A, B, C) based on marks obtained by a student.**
**if the percentage is above 90, assign grade A**
**if the percentage is above 75, assign grade B**
**if the percentage is above 65, assign grade C**

```
marks_total=int(input("\nEnter Total Marks:"))
marks_obtained=int(input("\nEnter Marks Obtained by Student:"))
percentage=(marks_obtained/marks_total)*100
if percentage>90:
    print("\nGrade:A\n\n",percentage,"%")
elif percentage>75:
    print("\nGrade:B\n\n",percentage,"%")
elif percentage>65:
    print("\nGrade:C\n\n",percentage,"%")
```



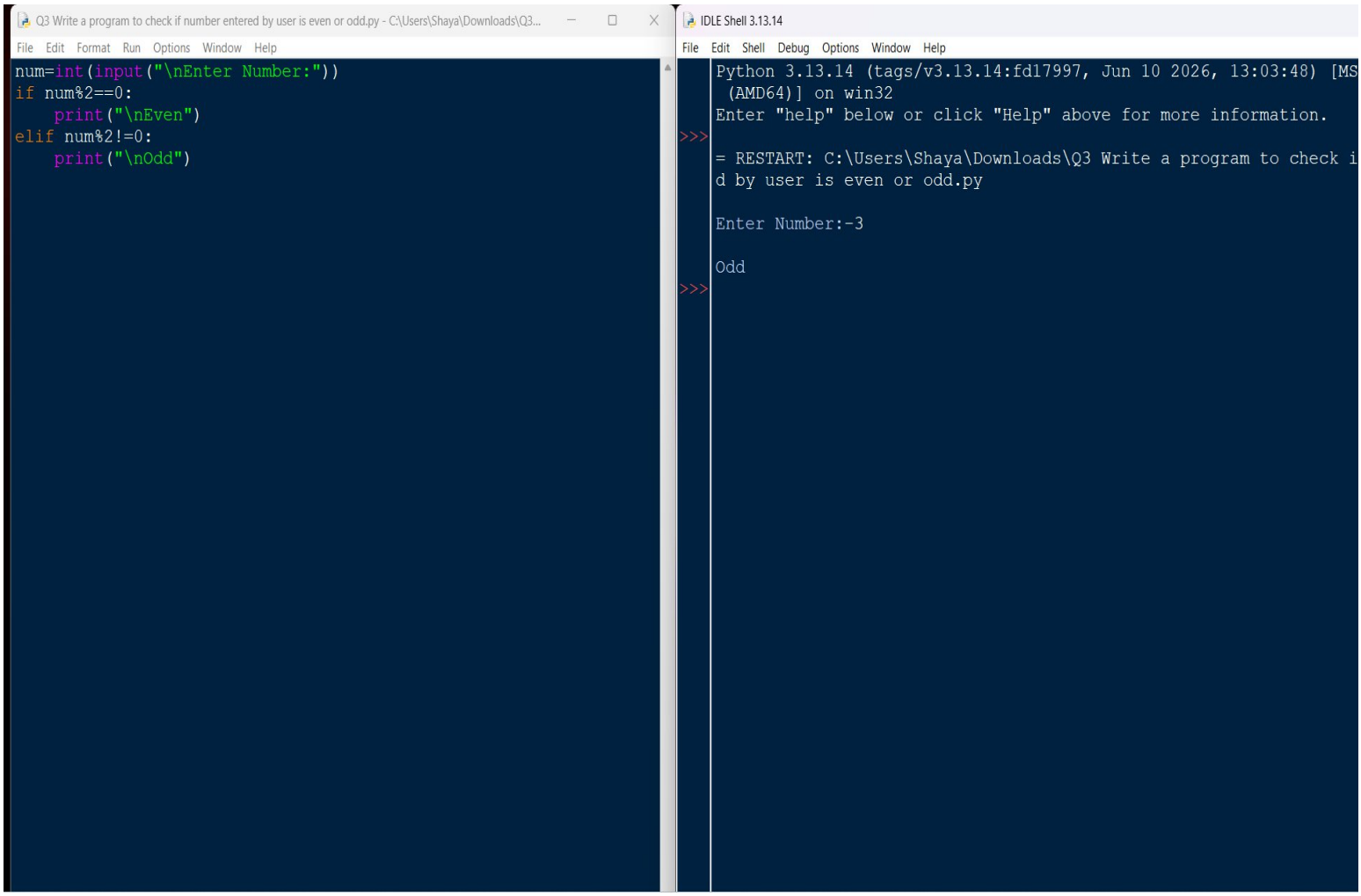
**Q2. Write a program to check if number entered by user is positive or negative.**

```
num=int(input("\nEnter Number:"))
if num>0:
    print("\nPositive")
elif num<0:
    print("\nNegative")
else:
    print("\nZero")
```



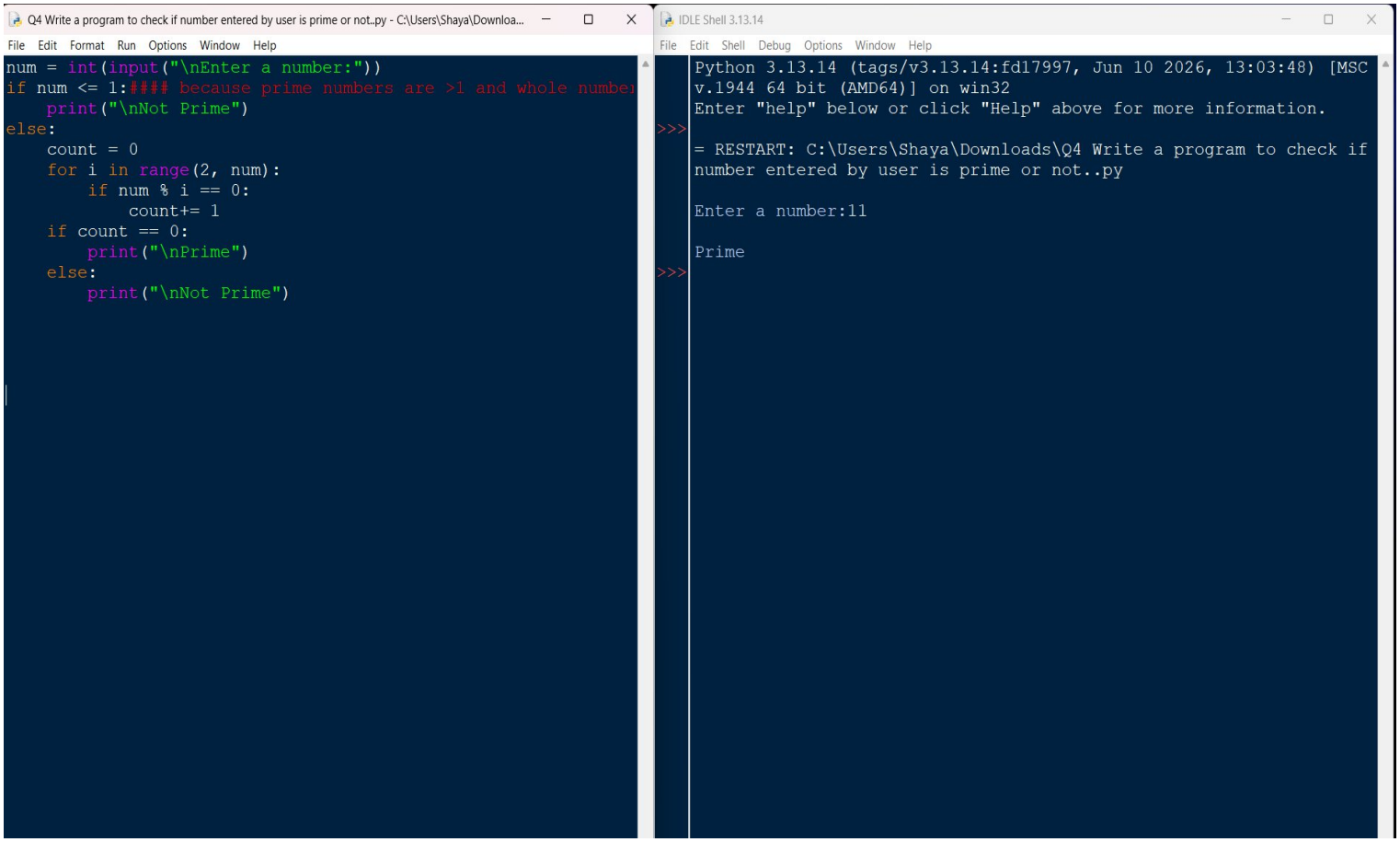
**Q3. Write a program to check if number entered by user is even or odd.**

```
num=int(input("\nEnter Number:"))
if num%2==0:
    print("\nEven")
elif num%2!=0:
    print("\nOdd")
```



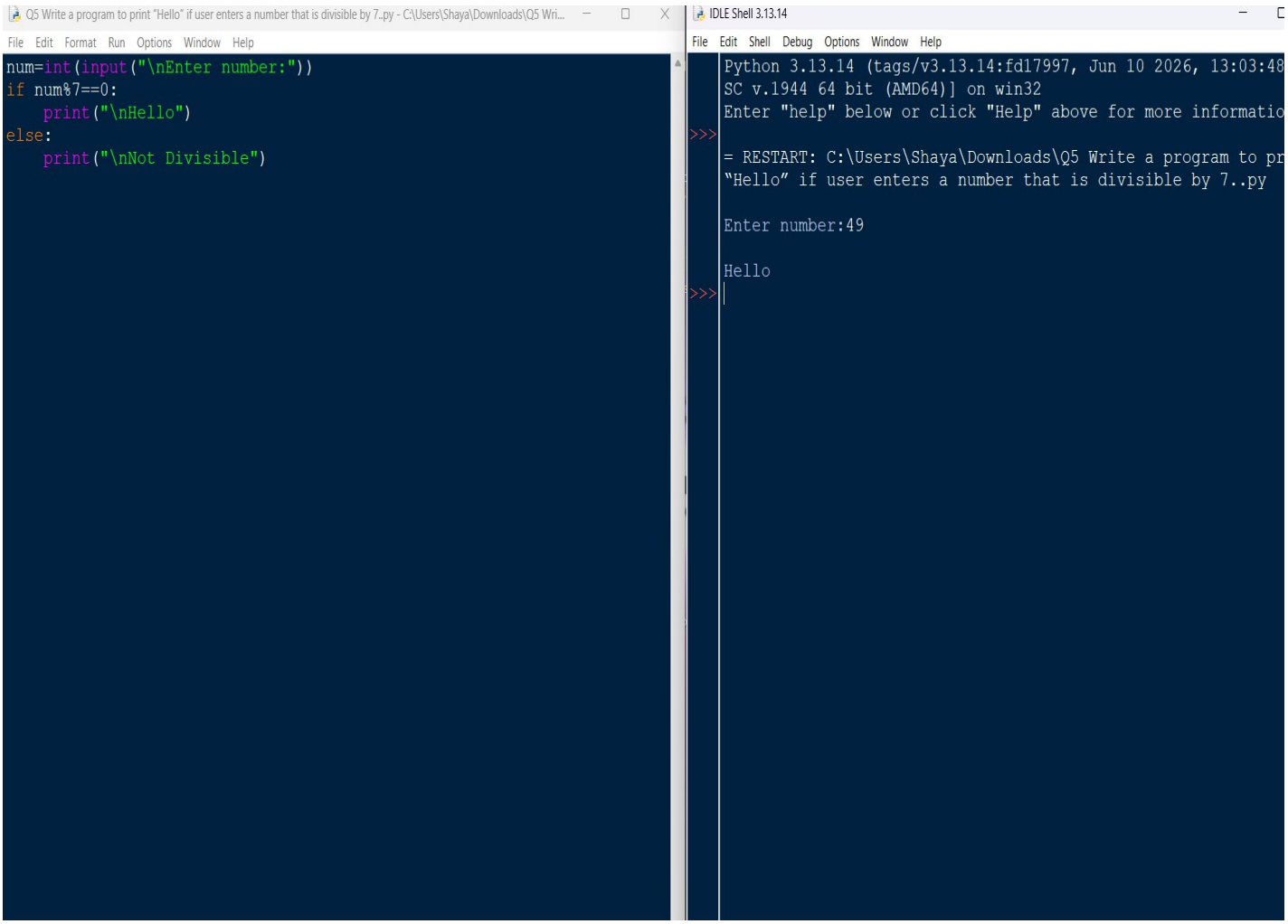
**Q4. Write a program to check if number entered by user is prime or not.**

```
num = int(input("\nEnter a number:"))
if num <= 1:
    print("\nNot Prime")
else:
    count = 0
    for i in range(2, num):
        if num % i == 0:
            count+= 1
    if count == 0:
        print("\nPrime")
    else:
        print("\nNot Prime")
```



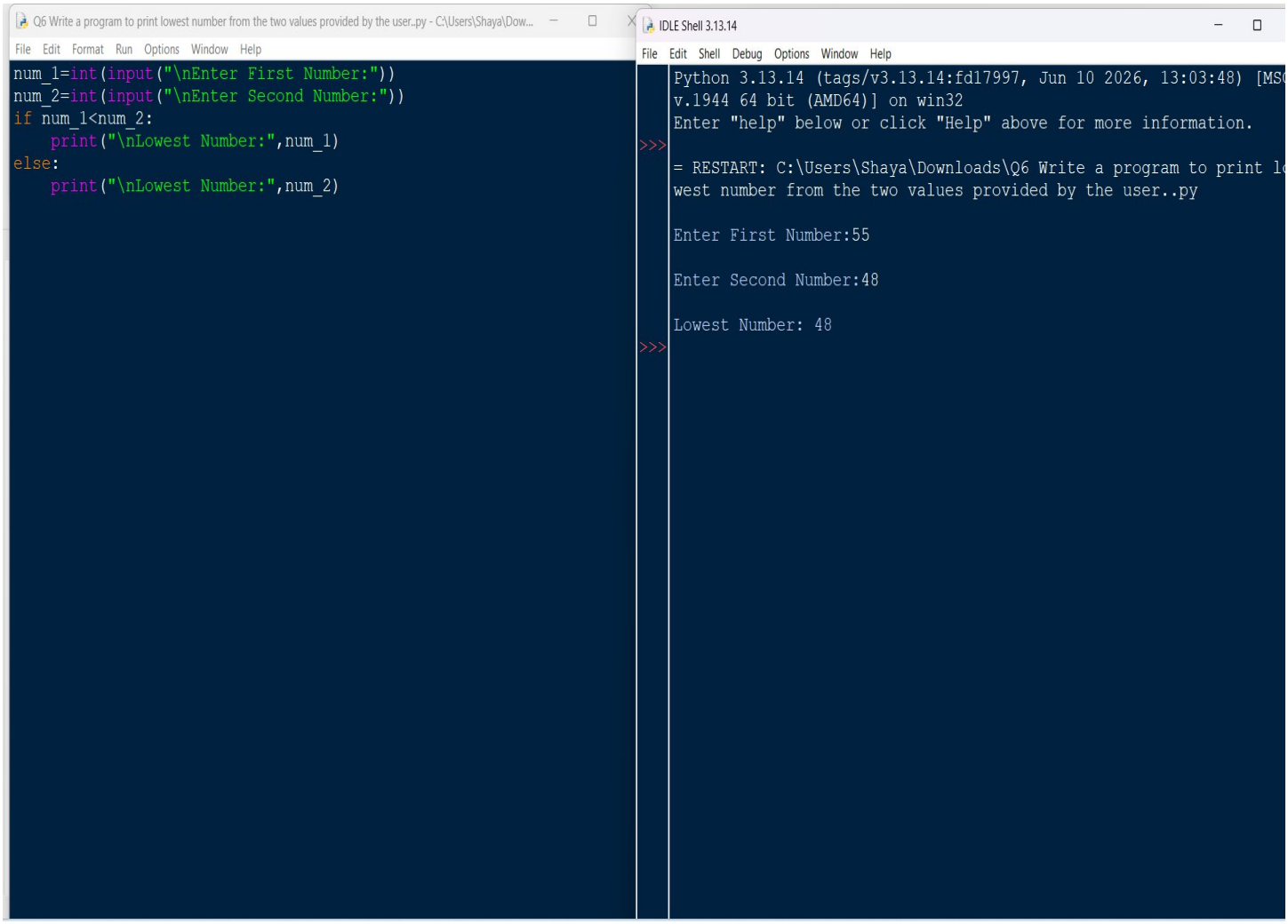
**Q5. Write a program to print “Hello” if user enters a number that is divisible by 7.**

```
num=int(input("\nEnter number:"))
if num%7==0:
    print("\nHello")
else:
    print("\nNot Divisible")
```



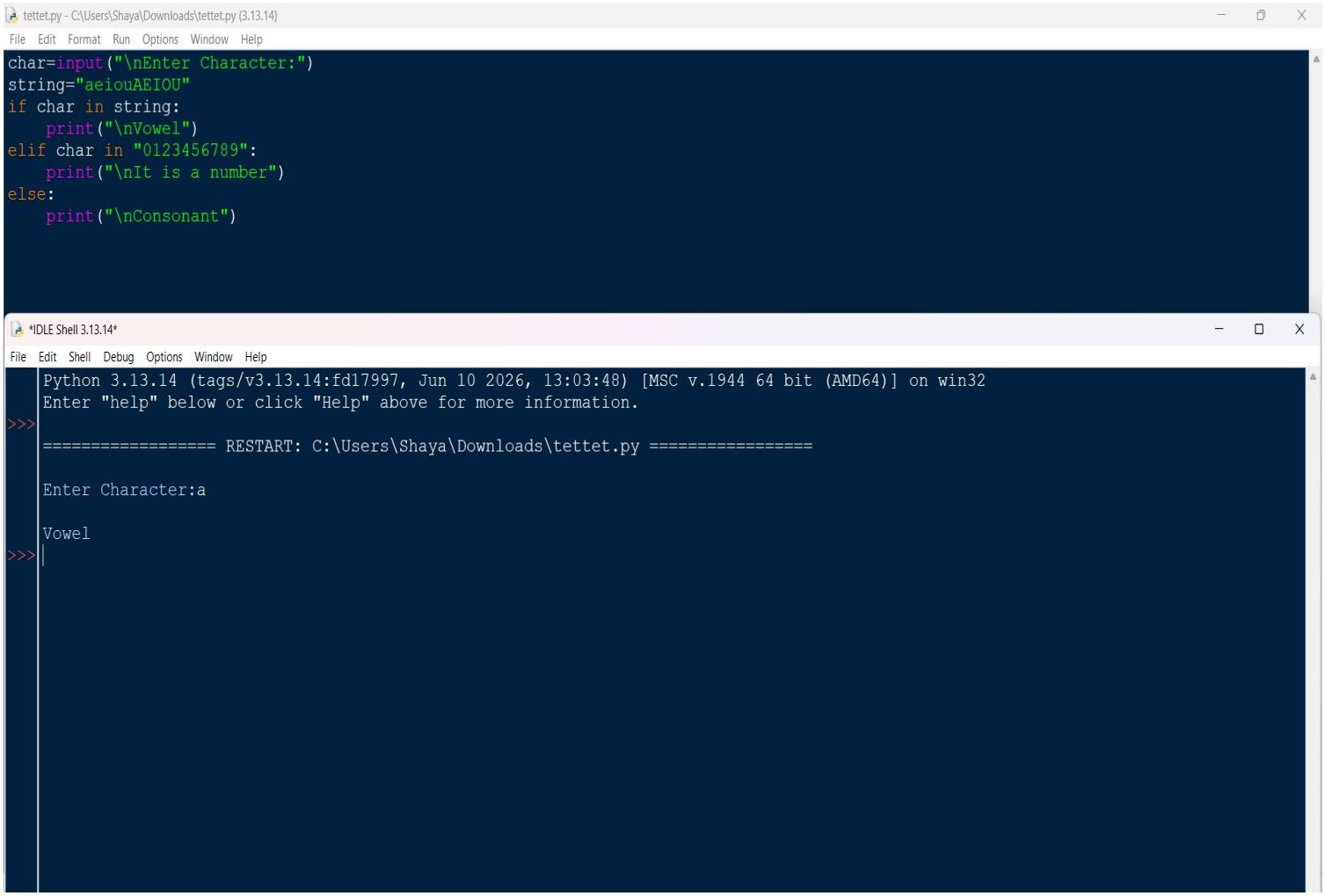
**Q6. Write a program to print lowest number from the two values provided by the user.**

```
num_1=int(input("\nEnter First Number:"))
num_2=int(input("\nEnter Second Number:"))
if num_1<num_2:
    print("\nLowest Number:",num_1)
else:
    print("\nLowest Number:",num_2)
```



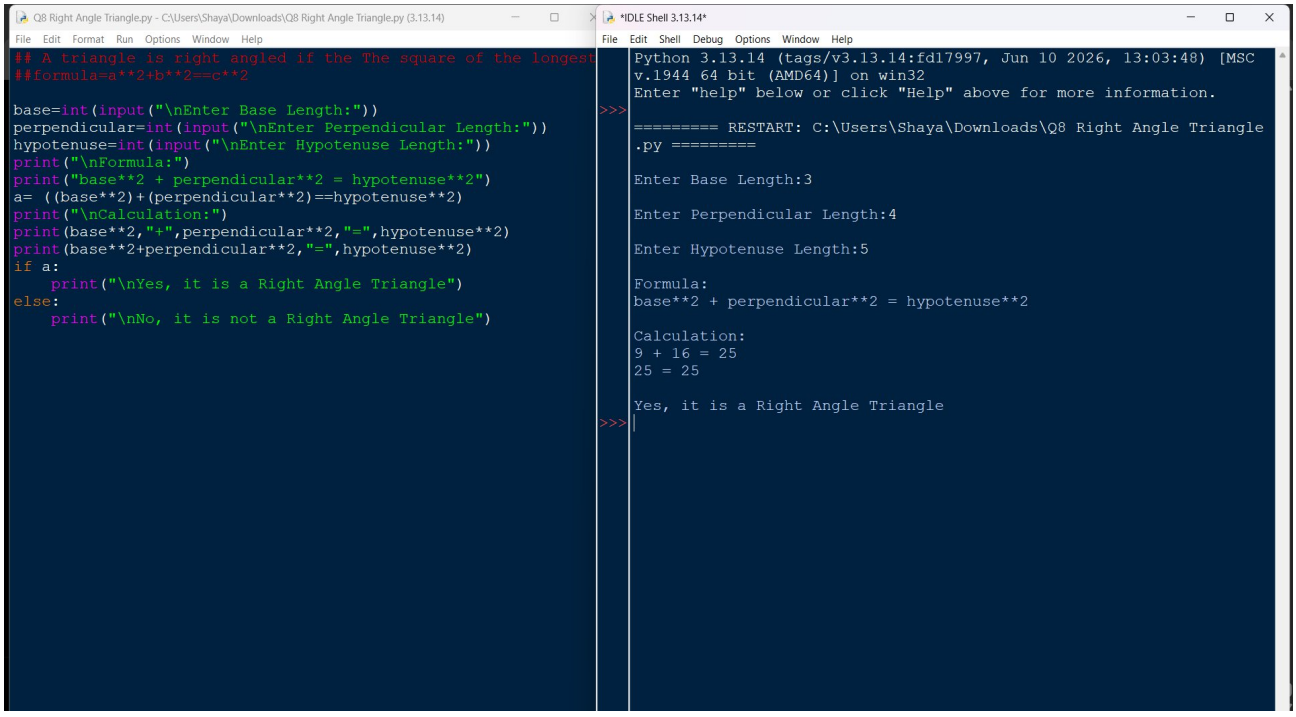
**Q7. Write a program to check if the character entered by user is vowel or consonant.**

```
char=input("\nEnter Character:")
string="aeiouAEIOU"
if char in string:
    print("\nVowel")
elif char in "0123456789":
    print("\nIt is a number")
else:
    print("\nConsonant")
```



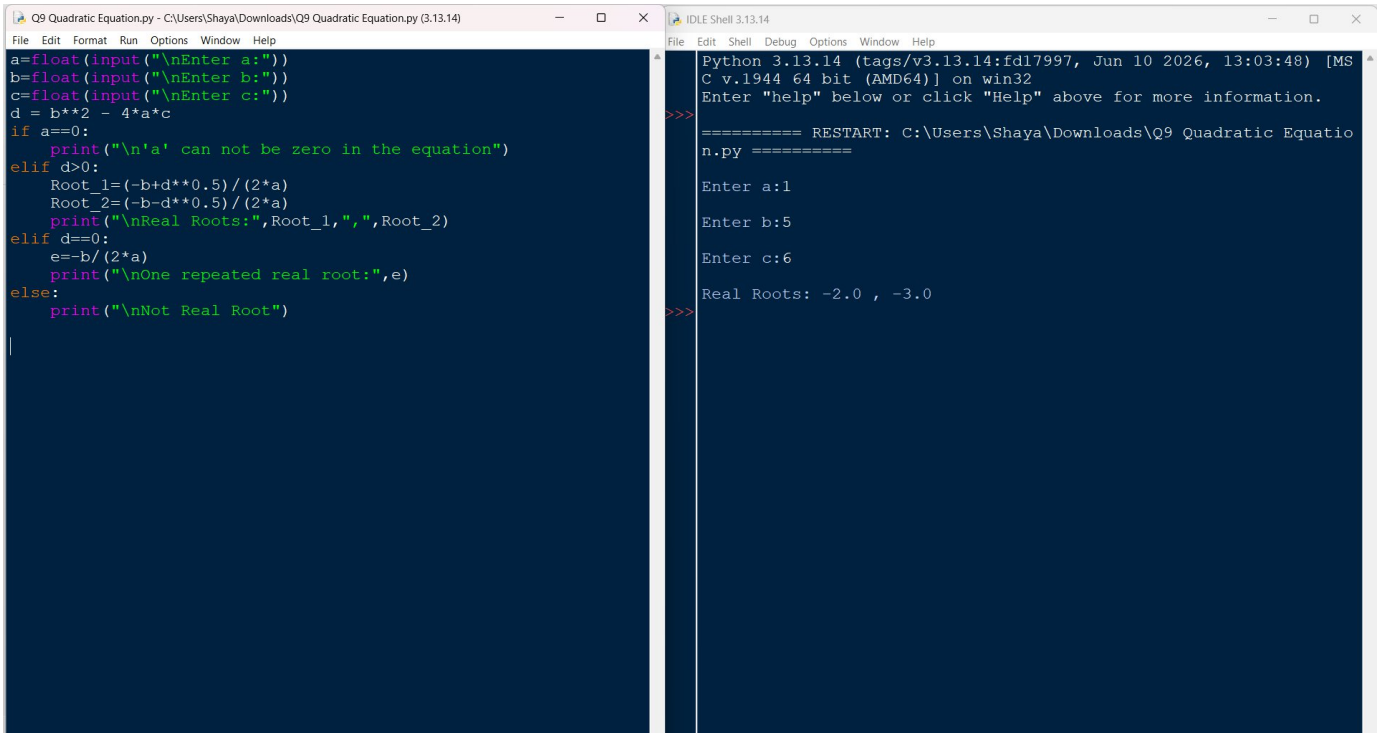
**Q8. Write a program that takes the dimensions (length of sides) of triangle to identify if the triangle is right angle triangle.**

```
base=int(input("\nEnter Base Length:"))
perpendicular=int(input("\nEnter Perpendicular Length:"))
hypotenuse=int(input("\nEnter Hypotenuse Length:"))
print("\nFormula:")
print("base**2 + perpendicular**2 = hypotenuse**2")
a= ((base**2)+(perpendicular**2)==hypotenuse**2)
print("\nCalculation:")
print(base**2,"+",perpendicular**2,"=",hypotenuse**2)
print(base**2+perpendicular**2,"=",hypotenuse**2)
if a:
    print("\nYes, it is a Right Angle Triangle")
else:
    print("\nNo, it is not a Right Angle Triangle")
```



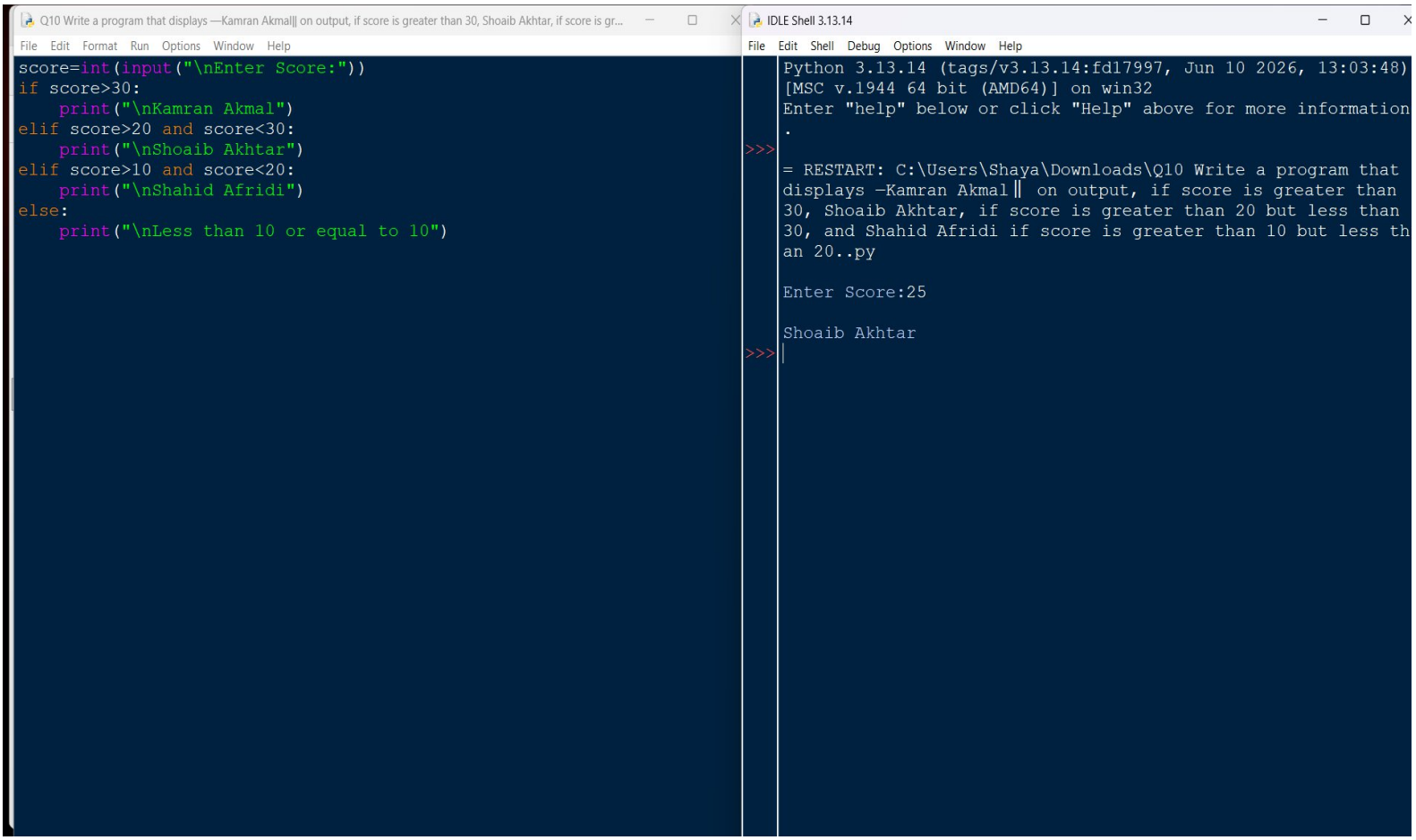
**Q9. Write a program that solves quadratic equation and prints the output only if the roots are real.**

```
a=float(input("\nEnter a:"))
b=float(input("\nEnter b:"))
c=float(input("\nEnter c:"))
d = b**2 - 4*a*c
if a==0:
    print("\n'a' can not be zero in the equation")
elif d>0:
    Root_1=(-b+d**0.5)/(2*a)
    Root_2=(-b-d**0.5)/(2*a)
    print("\nReal Roots:",Root_1,",",Root_2)
elif d==0:
    e=-b/(2*a)
    print("\nOne repeated real root:",e)
else:
    print("\nNot Real Root")
```



**Q10. Write a program that displays —Kamran Akmal‖ on output, if score >30, Shoaib Akhtar, if 20<score <30, and Shahid Afridi if 10<score <20.**

```
score=int(input("\nEnter Score:"))
if score>30:
    print("\nKamran Akmal")
elif score>20 and score<30:
    print("\nShoaib Akhtar")
elif score>10 and score<20:
    print("\nShahid Afridi")
else:
    print("\nLess than 10 or equal to 10")
```



**Q11. Write a program that takes password from user as input. Validate the password on the following criteria:**

**Password length between 7 to 15 characters which contain at least one numeric digit and a special character is acceptable.**

```
password = input("\nEnter your password to validate: ")

if len(password) < 7 or len(password) > 15:

    print("Invalid: Password must be between 7 and 15 characters
long")

elif("0" in password or "1" in password or "2" in password or
"3" in password or "4" in password or "5" in password or "6" in
password or "7" in password or "8" in password or "9" in
password) and ("!" in password or "@" in password or "#" in
password or "$" in password or "%" in password or "^" in
password or "&" in password or "*" in password or "(" in
password or ")" in password or "_" in password or "-" in
password or "+" in password or "=" in password or "{" in
password or "}" in password or "[" in password or "]" in
password or "|" in password or "\\" in password or ":" in
password or ";" in password or "\"" in password or "'" in
password or "<" in password or ">" in password or "," in
password or "." in password or "?" in password or "/" in
password or "~" in password or "`" in password):

    print("\nSuccess: Your password is valid and acceptable!")

else:

    print("\nInvalid: Password must contain at least one numeric
digit and one special character.")
```

Q11 password valudation.py - C:/Users/Shaya/Downloads/Q11 password valudation.py (3.13.14)

File Edit Format Run Options Window Help

```
password = input("\nEnter your password to validate: ")
if len(password) < 7 or len(password) > 15:
    print("Invalid: Password must be between 7 and 15 characters long")
elif("0" in password or "1" in password or "2" in password or "3" in
password or "4" in password or "5" in password or "6" in password or
"7" in password or "8" in password or "9" in password) and ("!" in
password or "@" in password or "#" in password or "$" in password or
"%" in password or "^" in password or "&" in password or "*" in
password or "(" in password or ")" in password or "_" in password or
"-" in password or "+" in password or "=" in password or "{" in
password or "}" in password or "[" in password or "]" in password or
"|" in password or "\\" in password or ":" in password or ";" in
password or "\"" in password or "'" in password or "<" in password or
">" in password or "," in password or "." in password or "?" in
password or "/" in password or "~" in password or "`" in password):
    print("\nSuccess: Your password is valid and acceptable!")
else:
    print("\nInvalid: Password must contain at least one numeric digit and one special character.")
```

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

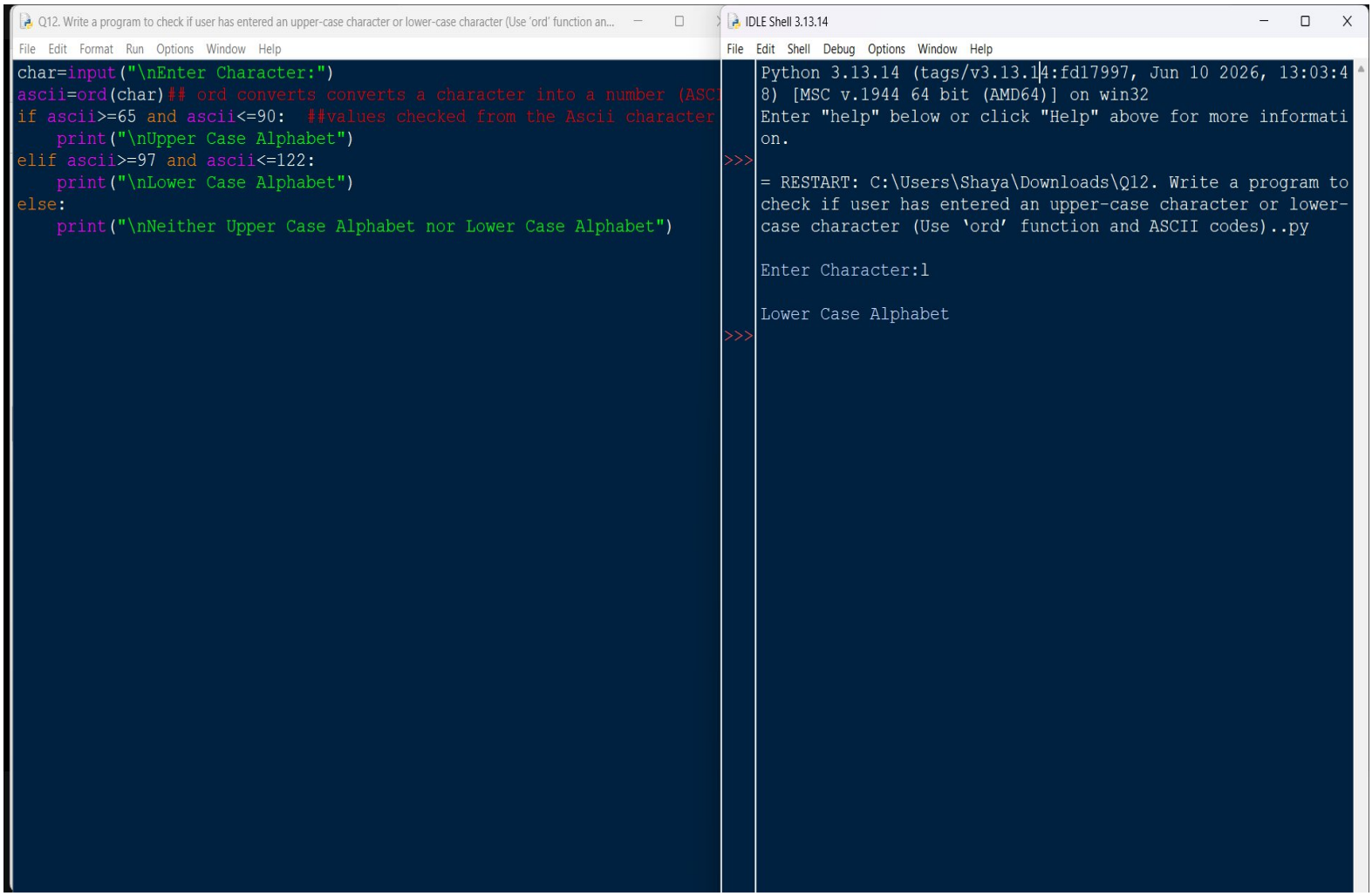
```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/Shaya/Downloads/Q11 password valudation.py ==========

Enter your password to validate: shayan123!

Success: Your password is valid and acceptable!
>>>
```

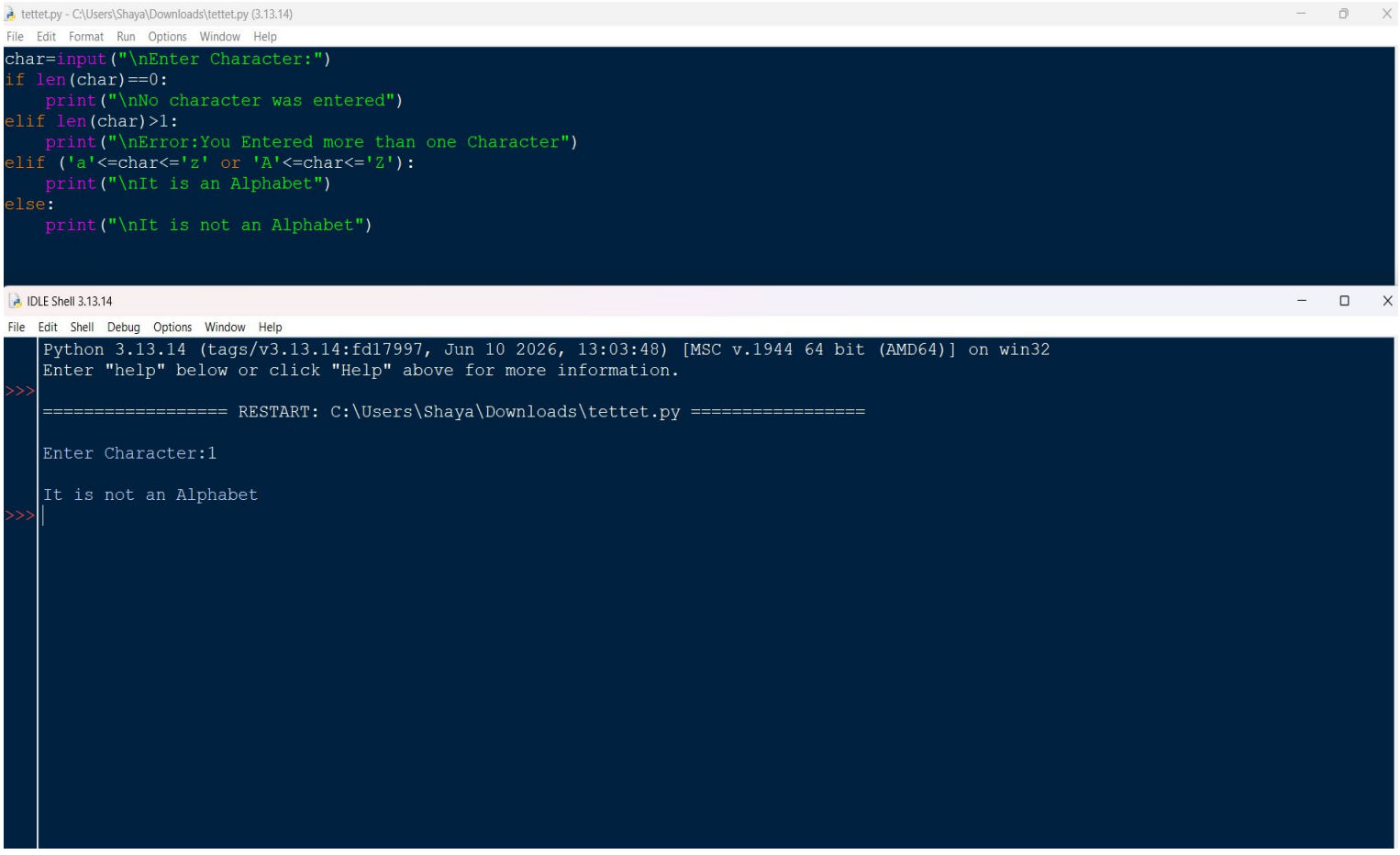
**Q12. Write a program to check if user has entered an upper-case character or lower-case character (Use 'ord' function and ASCII codes).**

```
char=input("\nEnter Character:")
ascii=ord(char)
if ascii>=65 and ascii<=90:
    print("\nUpper Case Alphabet")
elif ascii>=97 and ascii<=122:
    print("\nLower Case Alphabet")
else:
    print("\nNeither Upper Case Alphabet nor Lower Case
          Alphabet")
```



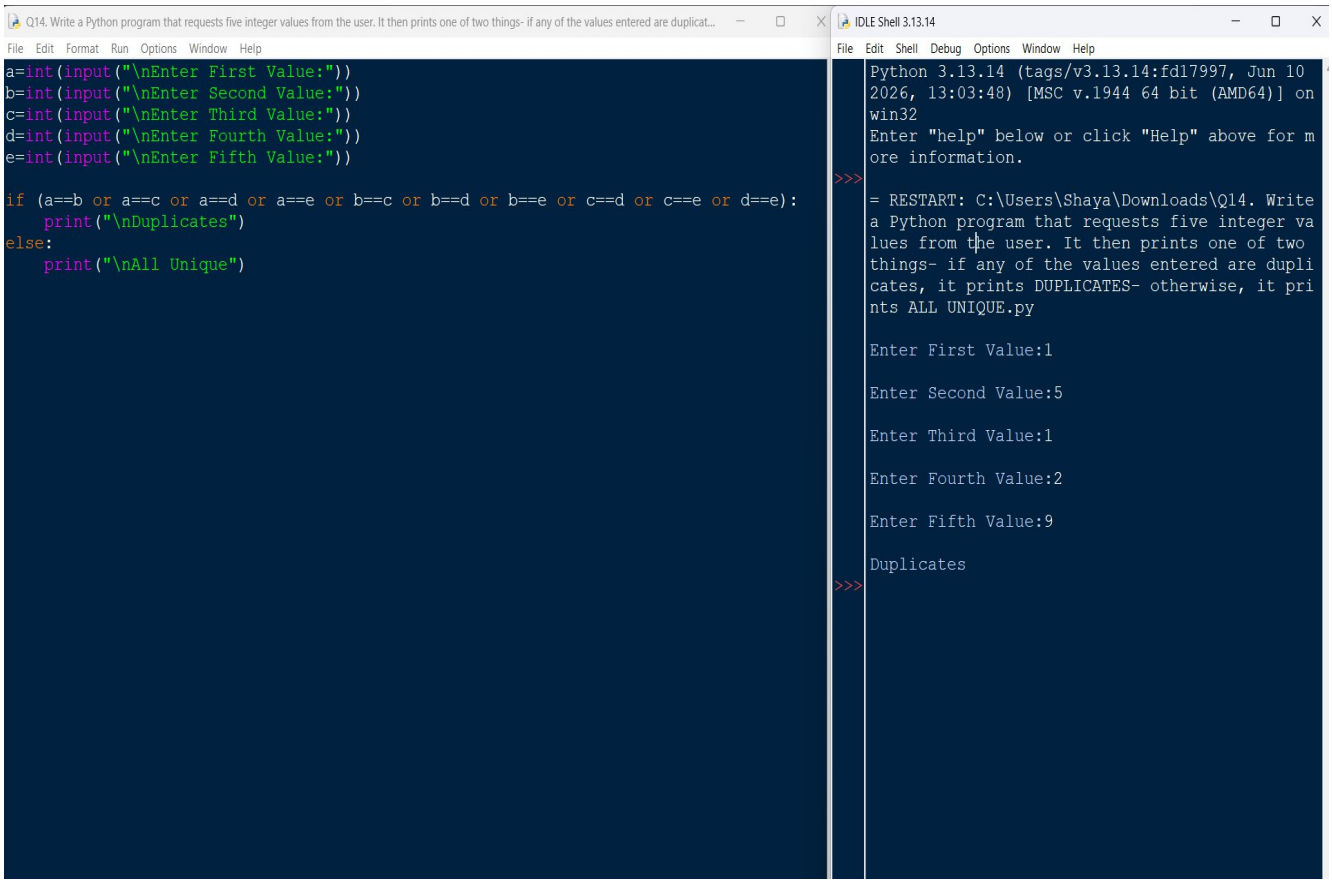
**Q13. Write a Python program to check if a character entered by the user is an alphabet or not. If the user enters more than one character as input, the program prints some appropriate error message and exit**

```
char=input("\nEnter Character:")
if len(char)==0:
    print("\nNo character was entered")
elif len(char)>1:
    print("\nError:You Entered more than one Character")
elif ('a'<=char<='z' or 'A'<=char<='Z'):
    print("\nIt is an Alphabet")
else:
    print("\nIt is not an Alphabet")
```



**Q14. Write a Python program that requests five integer values from the user. It then prints one of two things: if any of the values entered are duplicates, it prints "DUPLICATES"; otherwise, it prints "ALL UNIQUE".**

```
a=int(input("\nEnter First Value:"))
b=int(input("\nEnter Second Value:"))
c=int(input("\nEnter Third Value:"))
d=int(input("\nEnter Fourth Value:"))
e=int(input("\nEnter Fifth Value:"))
if (a==b or a==c or a==d or a==e or b==c or b==d or b==e or c==d or c==e or d==e):
    print("\nDuplicates")
else:
    print("\nAll Unique")
```



**Q15. Write a Python program that requests an integer value from the user. If the value is between 1 and 100 inclusive, print "OK"; otherwise, do not print anything**

```
num_1=int(input("\nEnter Value:"))
if 1<=num_1<=100:
    print("\nOK")
```

